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TWiki > Atlas Web > DataPreparation > LuminosityGroup > LuminosityForPhysics (2026-07-06, KlausMoenig)

LuminosityForPhysics

Introduction

[Luminosity for physics analysis](#)

[Luminosity requests](#)

[Useful Links and Documentation](#)

Proton-proton data

[Combination of 2022 to 2024 13.6 TeV proton-proton data \(preliminary\)](#)

[2024 13.6 TeV proton-proton \(preliminary\)](#)

[Combination of 2022 and 2023 13.6 TeV proton-proton data \(preliminary\)](#)

[2023 13.6 TeV proton-proton \(preliminary\)](#)

[2022 13.6 TeV proton-proton \(preliminary\)](#)

[2022 13.6 TeV proton-proton run for LHCf \(preliminary\)](#)

[2015 - 2018 13 TeV proton-proton \(final\)](#)

[Treatment of partial Run-2 datasets](#)

[Treatment of combinations of measurements](#)

[2017 and 2018 13 TeV proton-proton low-mu dataset \(final\)](#)

[2012 8 TeV proton-proton final](#)

[2011 7 TeV proton-proton final](#)

[2010 7 TeV proton-proton final](#)

Heavy Ion data

[2025 5.36 TeV neon-neon preliminary April 2026](#)

[2025 5.36 TeV oxygen-oxygen preliminary April 2026](#)

[2025 9.62 TeV proton-oxygen preliminary January 2026](#)

[2024 5.36 TeV proton-proton preliminary May 2026](#)

[2023 5.36 TeV lead-lead preliminary November 2025](#)

[2023 5.36 TeV lead-lead pre-preliminary July 2024](#)

[2018 5.02 TeV lead-lead final January 2022](#)

[2015 and 2017 5.02 TeV proton-proton \(final\)](#)

[2016 8.16 TeV proton-lead final January 2019](#)

[2015 5.0 TeV lead-lead final](#)

[2013 proton-lead final](#)

[2013 2.76 TeV proton-proton final](#)

[2011 2.76 TeV proton-proton final](#)

[2011 lead-lead \(2.76 TeV /nucleon\)](#)

[2010 lead-lead \(2.76 TeV /nucleon\)](#)

ALFA data

[2018 13 TeV proton-proton beta*=90m \(preliminary\)](#)

[2016 13 TeV proton-proton beta*=2.5km \(preliminary\)](#)

Introduction

This page describes the luminosity recommendations for physics analyses from the [LuminosityGroup](#). This includes the luminosity tag which is used to compute the integrated luminosity for a given dataset, and the associated systematic uncertainty to be propagated to the analysis results. The integrated luminosity is calculated separately by each analysis team, for its specific trigger and GRL setup. Instructions are available in the red box below.

Luminosity for physics analysis

Every analysis must calculate the luminosity for its specific setup, based on the luminosity tag, GRL, and trigger, using the [ATLAS luminosity calculator](#).

Please make sure to follow these steps:

- use the appropriate GRL, which you can find on the [GoodRunListsForAnalysis](#) page.
- use the recommended lumi tag, which you can find below.
- as "physics trigger" use the trigger used in your particular analysis.

- follow the other instructions on the lumicalc page.

Luminosity requests

If your specific dataset is missing on this page, please contact the luminosity conveners [✉atlas-luminosity-conveners](#). Luminosity requests are further handled through the [luminosity JIRA page](#). The JIRA ticket is created to make sure the information listed below is provided to allow the creation of a dedicated luminosity tag:

- A GRL with the list of runs and luminosity blocks which are good for analysis
- The luminosity algorithm that should be used for the luminosity calculation

The JIRA ticket should indicate explicitly the name of the data set for which the luminosity should be calculated and the requested timeline.

Useful Links and Documentation

- [ATLAS Luminosity Calculator](#)
- [Luminosity Calculator Tutorial](#)
- [LAr Event Veto Information](#)
- Information on available GRLs and where to download them can be found on the [GoodRunListsForAnalysis](#) page
- Previous, superseded recommendations can be found at the [LuminosityForPhysicsSuperseded](#) page

Proton-proton data

Combination of 2022 to 2024 13.6 TeV proton-proton data (preliminary)

Data Sample	data22/23/24_13p6TeV
Luminosity Uncertainty	1.9%
Documentation	ATL-DAPR-PUB-2025-001
Last update for this entry	13 November 2025

For any common analysis of 2022 to 2024 data just add the luminosities of the three years. The uncertainty of the combined luminosity is 1.9%. A brief explanation on the methodology, how this value is obtained can be found in [ATL-DAPR-PUB-2024-001](#). The reference for the PUB note is `\cite{ATL-DAPR-PUB-2025-001}` included in PubNotes.bib, the other references have not changed since Run-2 and can be found from the Run-2 entry: [LuminosityForPhysics#2015 2018 13 TeV proton proton f](#). The following sentence can be written in public CONF notes/papers: "The uncertainty in the integrated luminosity for data recorded in 2022-2024 is 1.9%\~ \cite{ATL-DAPR-PUB-2025-001}, following the methodology discussed in Ref.\~ \cite{DAPR-2021-01}, using the LUCID-2 detector~\cite{LUCID2} for the primary luminosity measurements, complemented by measurements using the inner detector and calorimeters."

2024 13.6 TeV proton-proton (preliminary)

Data Sample	data24_13p6TeV
Luminosity Tag	OfiLumi-Run3-008
Luminosity Uncertainty	2.0%
Documentation	ATL-DAPR-PUB-2025-001
Additional Information	vdM analysis , vdM and combination , mu corrections , calibration transfer
Last update for this entry	13 November 2025

The preliminary calibration of 2024 13.6 TeV pp data is available in the tag OfiLumi-Run3-008. This result reduces the luminosity recorded by ATLAS by around 1.2% compared to the OfiLumi-Run3-004 tag. The luminosity is constantly 0.26% higher than in the OfiLumi-Run3-007 tag.

Combination of 2022 and 2023 13.6 TeV proton-proton data (preliminary)

Data Sample	data22/23_13p6TeV
Luminosity Uncertainty	2.0%
Documentation	ATL-DAPR-PUB-2024-001
Last update for this entry	27 August 2024

For any common analysis of 2022 and 2023 data just add the luminosities of the two years. The uncertainty of the combined luminosity is 2.0%. A brief explanation how this value is obtained can be found in [ATL-DAPR-PUB-2024-001](#).

2023 13.6 TeV proton-proton (preliminary)

Data Sample	data23_13p6TeV
Luminosity Tag	OfLumi-Run3-004
Luminosity Uncertainty	2.0%
Documentation	ATL-DAPR-PUB-2024-001
Additional Information	sigma_vis Calibration Linearity and long-term stability Calibration Transfer Tag Validation
Last update for this entry	25 August 2024

The preliminary calibration of 2023 13.6 TeV pp data is available in the tag OfLumi-Run3-004. This result reduces the luminosity recorded by ATLAS by around 0.7% compared to the previous OfLumi-Run3-003 tag. Figures and tables documenting the calibration are published in [ATL-DAPR-PUB-2024-001](#).

2022 13.6 TeV proton-proton (preliminary)

Data Sample	data22_13p6TeV
Luminosity Tag	OfLumi-Run3-003
Luminosity Uncertainty	2.2%
Documentation	ATL-DAPR-PUB-2023-001
Additional Information	sigma_vis calibration , mu calibration and tag presentation , Combined uncertainties
Last update for this entry	16 February 2023

The preliminary calibration of 2022 13.6TeV pp data uses a calibration based on the vdM scans recorded 10-11 November 2022 (run 439428), as well as a mu-correction determined in a standard physics fill at high mu on 20 August 2022 (run 431885) based on track counting data, which is anchored to LUCID in the quiescent periods of the vdM run above. The default luminometer from the LUCID side is LUCID Bi2 HITOR. Figures and tables documenting the calibration are published in [ATL-DAPR-PUB-2023-001](#).

The updated tag shifts the integrated luminosity down by -5.4%. Analyses should re-evaluate the integrated luminosity of their GRL using the [LumiCalc tool](#). The following sentence can be written in public CONF notes/papers: "The uncertainty in the integrated luminosity for data recorded in 2022 is 2.2%~\cite{ATL-DAPR-PUB-2023-001}, following the methodology discussed in Ref.~\cite{DAPR-2021-01}, using the LUCID-2 detector~\cite{LUCID2} for the primary luminosity measurements, complemented by measurements using the inner detector and calorimeters."

The reference for the PUB note is \cite{ATL-DAPR-PUB-2023-001} included in [PubNotes.bib](#), the other references have not changed since Run-2 and can be found from the Run-2 entry: [LuminosityForPhysics#2015_2018_13_TeV_proton_proton_f](#).

2022 13.6 TeV proton-proton run for LHCf (preliminary)

Data Sample	data22_13p6TeV
Luminosity Tag	OfLumi-Run3-003
Luminosity Uncertainty	1.5%
Documentation	ATL-DAPR-PUB-2023-001
Additional Information	sigma_vis calibration , Time stability checks
Last update for this entry	9 September 2025

The preliminary calibration of 2022 13.6TeV pp LHCf data uses a calibration based on the vdM scans recorded 10-11 November 2022 (run 439428), as well as stability tests based on tracking and LUCID. The default luminometer from the LUCID side is LUCID Bi2 HITOR. Figures and tables documenting the vdM calibration are included in [ATL-DAPR-PUB-2023-001](#).

The following sentence can be written in public CONF notes/papers: "The uncertainty in the integrated luminosity for the LHCf data recorded in 2022 is 1.5%. The error is composed of 1.45% from the vdM calibration~\cite{ATL-DAPR-PUB-2023-001}, following the methodology discussed in Ref.~\cite{DAPR-2021-01}, using the LUCID-2 detector~\cite{LUCID2} for the primary luminosity measurements. An uncertainty of 0.5% is added in quadrature due to the extrapolation of the LUCID measurement from the vdM calibration run to the LHCf run, checked by comparing LUCD to tracking."

2015 - 2018 13 TeV proton-proton (final)

Data Sample	data15/16/17/18_13TeV
Luminosity Tag	OfLumi-13TeV-011
Luminosity Uncertainty	0.83%
Documentation	Eur. Phys. J. C 83, 982 (2023) Tag validation , Luminosity group approval with preliminary beam-beam uncertainty , Final results with updated beam-beam uncertainty
Last update for this entry	30 January 2023

The final calibration of the 2015-2018 13 TeV pp data sample is now available in the tag OfLumi-13TeV-011. The uncertainty on the combined data set (2015-2018) is reduced to 0.83% from 1.7% previously [[ATLAS-CONF-2019-021](#)].

Please note that this tag shifts the integrated luminosity up by about +0.8%, and the shift itself is not a constant scaling factor throughout the year(s). Analyses should therefore re-evaluate the integrated luminosity of their GRL using the [LumiCalc](#) tool. A rerunning of the pileup reweighting is not needed - you can continue to use the files from tag OfLumi-13TeV-010 for this. Note: The specified uncertainty is only valid for the full Run-2 dataset. If you use only a subset or individual years, please check out recommendations for partial Run-2 combinations below, and please get in contact with the luminosity conveners, if your combination is not yet listed.

The tag is also applicable for the low mu dataset in 2017 and 2018, with a different uncertainty though - see the [separate entry](#) for details. It does not include the high β^* special ALFA datasets.

The following sentence can be written in public CONF notes/papers: "The uncertainty in the combined 2015–2018 integrated luminosity is 0.83%\ \cite{DAPR-2021-01}, obtained using the LUCID-2 detector \cite{LUCID2} for the primary luminosity measurements, complemented by measurements using the inner detector and calorimeters."

Please quote the total luminosity as follows:

- In figure legends and in text if the uncertainty is not quoted: '140 fb⁻¹'
- If the absolute uncertainty is meant to be quoted: '140.1 ± 1.2 fb⁻¹', following PDG rounding recommendations

The value of 140 fb⁻¹ corresponds to the standard 'AllGood' physics GRL; if you use a different GRL, the integrated luminosity may be different.

Cite:

- [DAPR-2021-01] \cite{DAPR-2021-01} from ATLAS.bib
- [LUCID2] \cite{LUCID2} from ATLAS.bib

```
@Article{DAPR-2021-01,
  author      = "{ATLAS Collaboration}",
  title       = "{Luminosity determination in $pp$ collisions at $\sqrt{s}=13$ TeV using the ATLAS detector at the LHC}",
  eprint      = "2212.09379",
  archivePrefix = "arXiv",
  primaryClass = "hep-ex",
  reportNumber = "CERN-EP-2022-281",
  doi = "10.1140/epjc/s10052-023-11747-w",
  journal     = "Eur. Phys. J. C",
  volume      = "83",
  number      = "10",
  pages       = "982",
  year        = "2023"
}
```

Treatment of partial Run-2 datasets

If you use only a partial Run-2 dataset, please check the available combinations below, and get in contact with the luminosity conveners: atlas-luminosity-conveners@cern.ch, if the one you need is missing.

For the combination of 2015+2016 data, using the same GRLs as for full Run-2 above, the luminosity information is given below:

Data Sample	data15+16_13TeV
Luminosity Uncertainty	0.87%
Documentation	Partial Run-2 combination

The following sentence for the 2015+2016 partial Run-2 data can be written in public CONF notes/papers: "The uncertainty in the integrated luminosity of the combined dataset from 2015 and 2016 is 0.87%, using the same methodology as in Ref. \cite{DAPR-2021-01}, and obtained using the LUCID-2 detector \cite{LUCID2} for the primary luminosity measurements, complemented by measurements using the inner detector and calorimeters."

Please quote as follows:

- In figure legends and in text if the uncertainty is not quoted: ' 37 fb^{-1} '
- If the absolute uncertainty is meant to be quoted: ' $36.65 \pm 0.32 \text{ fb}^{-1}$ ', following PDG rounding recommendations

The references remain as for full Run-2.

For the combination of 2016+2017+2018 data, using the same GRLs as for full Run-2 above, the luminosity information is given below:

Data Sample	data16+17+18_13TeV
Luminosity Uncertainty	0.84%
Documentation	Partial Run-2 combination - Part 2

The following sentence for the 2016+2017+2018 partial Run-2 data can be written in public CONF notes/papers: "The uncertainty in the integrated luminosity of the combined dataset from 2016, 2017 and 2018 is 0.84%, using the same methodology as in Ref. \cite{DAPR-2021-01}, and obtained using the LUCID-2 detector \cite{LUCID2} for the primary luminosity measurements, complemented by measurements using the inner detector and calorimeters."

Please quote as follows:

- In figure legends and in text if the uncertainty is not quoted: ' 137 fb^{-1} '
- If the absolute uncertainty is meant to be quoted: ' $136.8 \pm 1.1 \text{ fb}^{-1}$ ', following PDG rounding recommendations

The references remain as for full Run-2.

Treatment of combinations of measurements

Instructions for combinations of analyses using preliminary and final luminosity calibrations, as well as using full and partial datasets can be found in this presentation: [Combination instructions, 17 Aug 2023](#)

If you intend to combine with Run-1 measurements, please contact the luminosity conveners (see below) early in your analysis process.

In case something is not clear, or you are interested in another case of combinations, please get in contact with the luminosity conveners: atlas-luminosity-conveners@cern.ch.

2017 and 2018 13 TeV proton-proton low-mu dataset (final)

Data Sample	data17/18_13TeV (low mu)
Luminosity Tag	OfiLumi-13TeV-011
Luminosity Uncertainty	0.92%
Documentation	Tag validation , Luminosity group approval with preliminary beam-beam uncertainty , Final results with updated beam-beam uncertainty
Last update for this entry	20 December 2022

The final calibration of the 2017 and 2018 13 TeV pp low-mu data sample is now available in the tag OfiLumi-13TeV-011. The luminosity uncertainty for the combined dedicated low-mu dataset in 2017 and 2018 is reduced to 0.92% from 1.5% previously [[ATLAS-CONF-2020-](#)

[023](#). Note: The specified uncertainty is only valid for the combined 2017-2018 low-mu dataset. If you use only a subset or one of the individual years, please get in contact with the luminosity conveners.

For the GRLs, please check the DQ page: [GoodRunListsForAnalysisRun2#Low_mu_proton_proton_data_taking](#)

The following sentence can be written in public CONF notes/papers: "The uncertainty in the integrated luminosity of the dedicated dataset in 2017 and 2018 used in this analysis is 0.92% \cite{DAPR-2021-01}, obtained using the LUCID-2 detector \cite{LUCID2} for the primary luminosity measurements, complemented by measurements using the inner detector and calorimeters."

Please quote as follows:

- In figure legends and in text if the uncertainty is not quoted: '338 pb⁻¹'
- If the absolute uncertainty is meant to be quoted: '338.1 ± 3.1 pb⁻¹', following PDG rounding recommendations

The value of 338 pb⁻¹ corresponds to the standard 'AllGood' lowMu physics GRL; if you use a different GRL, the integrated luminosity may be different.

Cite:

- [DAPR-2021-01] See [LuminosityForPhysics#2015_2018_13_TeV_proton_proton_f](#)
- [LUCID2] as previously - see [LuminosityForPhysics#2018_13_TeV_proton_proton_Morion](#)

2012 8 TeV proton-proton final

Data Sample	data12_8TeV
Luminosity Tag	OfLumi-8TeV-004
Lumi Uncertainty	1.9%
Documentation	Eur. Phys. J. C 76 (2016) 653
Last update for this entry	22 Feb 2017

The following sentence(s) can be written in public conf notes/paper: "The absolute luminosity scale is derived from beam-separation scans performed in November 2012. The uncertainty in the integrated luminosity is 1.9%~\cite{DAPR-2013-01}."

The luminosity uncertainties at 8 TeV (2012) and at 7 TeV (2010+2011) should be treated as uncorrelated, for the following reasons.

- The vdM-calibration uncertainties in 2011 & 2012 are dominated by beam conditions at the time of the scan. These vary from scan session to scan session, even at the same c.m. energy. More information on this topic can be found in W. Kozanecki's presentation to the 2012 Lumi Days workshop: <https://indico.cern.ch/event/162948/session/5/contribution/18/material/0/>
- The vdM-calibration uncertainties in 2010 are dominated by the bunch-current product. These are more than 5 times larger in 2010 than in 2011 & 2012, and their origin is such that the 2011/12 bunch-population uncertainty is uncorrelated with the corresponding 2010 uncertainty.
- In addition, the total uncertainty in the 2012 integrated luminosity receives large contributions from instrumental effects (long-term stability, calibration transfer from vdM to physics conditions) that did not affect the 2010 & 2011 luminosity measurements. A tabulated comparison of the 2011 and 2012 luminosity uncertainties can be found here: https://indico.cern.ch/event/577659/contributions/2339494/attachments/1355174/2047736/Run1_L_2011n2012_systs_WK_14Oct16.pdf

The correlations between the ATLAS & CMS luminosity uncertainties at 7 and 8 TeV were finalized, and agreed upon between ATLAS & CMS, during the LLCMWG meeting of 6 Feb 2017. The conclusions of this meeting are documented in this LMTF presentation of 16 Feb 2017: https://indico.cern.ch/event/571436/contributions/2479379/attachments/1413751/2169558/ATLAS-CMS_crltnsUpd_R1_WK_6n16Feb17.pdf, where pages 15 & 16 summarize in tabular form the correlations between ATLAS & CMS luminosity uncertainties applicable to the 2011 and 2012 running periods.

[DAPR-2013-01] ATLAS Collaboration, Luminosity Determination in pp Collisions at sqrt(s) = 8 TeV using the ATLAS Detector at the LHC, Eur. Phys. J. C 76 (2016) 653, arXiv:1608.03953 [hep-ex].

2011 7 TeV proton-proton final

Data Sample	data11_7TeV
Luminosity Tag	OfLumi-7TeV-004
Lumi Uncertainty	1.8%
Documentation	Paper accepted by EPJC

The luminosity scale has been modified by a constant factor of $1/1.0141 = 0.98610$ from OfiLumi-7TeV-003 to account for beam-beam effects during the vdM scans. The uncertainty is unchanged from the previous tag.

The 2010 & 2011 luminosity uncertainties should be treated as uncorrelated, for the following reasons.

- The vdM-calibration uncertainties in 2011 are dominated by beam conditions at the time of the scan. These vary from scan session to scan session, even at the same c.m. energy. More information on this topic can be found in W. Kozanecki's presentation to the 2012 Lumi Days workshop: <https://indico.cern.ch/event/162948/session/5/contribution/18/material/0/>
- The vdM-calibration uncertainties in 2010 are dominated by the bunch-current product. These are more than 5 times larger in 2010 than in 2011, and their origin is such that the 2011 bunch-population uncertainty is uncorrelated with the corresponding 2010 uncertainty.

2010 7 TeV proton-proton final

Data Sample	data10_7TeV
Luminosity Tag	OfiLumi-7TeV-004
Lumi Uncertainty	3.5%
Documentation	Paper accepted by EPJC

The luminosity scale has been modified by a constant factor of $1/1.0187 = 0.98164$ from OfiLumi-7TeV-003 to account for beam-beam effects during the vdM scans. The uncertainty has slightly increased (from 3.4%) compared to the previous tag.

Heavy Ion data

2025 5.36 TeV neon-neon preliminary April 2026

Data Sample	data25_hi
Luminosity Tag	OfiLumi-Run3-009
Lumi Uncertainty	4.6%
Documentation	Slides 16/04/2026

This is the preliminary calibration for the 2025 Ne+Ne 5.36 TeV/n data with a total uncertainty of 4.6%. The following sentences can be written in public CONF notes: "The uncertainty in the integrated luminosity of the data sample is 4.6%. It is derived from the calibration of the luminosity scale using $\sqrt{s}$ - $\sqrt{s}$ emittance scans, following a methodology similar to that detailed in Ref.~\cite{DAPR-2021-01}, and using the LUCID-2 detector for the baseline luminosity measurements \cite{LUCID2}."

2025 5.36 TeV oxygen-oxygen preliminary April 2026

Data Sample	data25_hi
Luminosity Tag	OfiLumi-Run3-009
Lumi Uncertainty	2.6%
Documentation	Slides 23/04/2026

This is the preliminary calibration for the 2025 O+O 5.36 TeV/n data with a total uncertainty of 2.6%. The following sentences can be written in public CONF notes: "The uncertainty in the integrated luminosity of the data sample is 2.6%. It is derived from the calibration of the luminosity scale using $\sqrt{x}$ - $\sqrt{y}$ beam-separation scans, following a methodology similar to that detailed in Ref.~\cite{DAPR-2021-01}, and using the LUCID-2 detector for the baseline luminosity measurements \cite{LUCID2}."

2025 9.62 TeV proton-oxygen preliminary January 2026

Data Sample	data25_hi
Luminosity Tag	OfiLumi-Run3-009
Lumi Uncertainty	2.3%
Documentation	Slides 15/01/26

This is the preliminary calibration for the 2025 p+O 9.62 TeV/n data with a total uncertainty of 2.3%. The following sentences can be written in public CONF notes: "The uncertainty in the integrated luminosity of the data sample is 2.3%. It is derived from the calibration of the

2024 5.36 TeV proton-proton preliminary May 2026

Data Sample	data24_5p36TeV
Luminosity Tag	OfiLumi-Run3-009
Luminosity Uncertainty	1.5%
Documentation	Slides: vdM calibration Calibration transfer and long-term stability

This is the preliminary calibration for the 2025 Pb+Pb 5.36 TeV/n data with a total uncertainty of 1.5%. The following sentence can be written in public CONF notes: "The uncertainty in the integrated luminosity of the data sample is 1.5%. It is derived from the calibration of the luminosity scale using x - y beam-separation scans, following a methodology similar to that detailed in Ref.~\cite{DAPR-2021-01}, and using the LUCID-2 detector for the baseline luminosity measurements \cite{LUCID2}, complemented by measurements using the inner detector and calorimeters."

2023 5.36 TeV lead-lead preliminary November 2025

Data Sample	data25_hi
Luminosity Tag	OfiLumi-Run3-009
Lumi Uncertainty	2.6%
Documentation	Note ATL-COM-DAPR-2025-021

This is the preliminary calibration for the 2023 Pb+Pb 5.36 TeV/n data with a total uncertainty of 2.6%. The luminosity went up by 2% compared to the pre-preliminary calibration. The following sentences can be written in public CONF notes: "The uncertainty in the integrated luminosity of the data sample is 2.6%. It is derived from the calibration of the luminosity scale using x - y beam-separation scans, following a methodology similar to that detailed in Ref.~\cite{DAPR-2021-01}, and using the LUCID-2 detector for the baseline luminosity measurements \cite{LUCID2}."

2023 5.36 TeV lead-lead pre-preliminary July 2024

Data Sample	data23_hi
Luminosity Tag	OfiLumi-Run3-005
Lumi Uncertainty	3.5%
Documentation	Slides 18/07/24

This is the first pre-preliminary calibration for the 2023 Pb+Pb 5.36 TeV/n data that was produced to support the heavy-ion monopole analysis that was presented at ICHEP24. The uncertainty of 3.5% reflects the pre-preliminary nature of this result, where several steps of the vdM analysis were either not performed, or done with a very simple treatment to get a first understanding. In particular, the length scale analysis and orbit drift corrections were not available.

The following sentences can be written in public CONF notes: "The uncertainty in the integrated luminosity of the data sample is 3.5%. It is derived from the calibration of the luminosity scale using x - y beam-separation scans, following a methodology similar to that detailed in Ref.~\cite{DAPR-2021-01}, and using the LUCID-2 detector for the baseline luminosity measurements \cite{LUCID2}." If you plan to release a conf note or publish a paper on this dataset, please contact the luminosity conveners, as an improved value may be possible on a short timescale.

2018 5.02 TeV lead-lead final January 2022

Data Sample	data18_hi
Luminosity Tag	OfiLumi-HI-010
Lumi Uncertainty	2.0% or 1.9%
Documentation	Slides 16/12/21 and 13/1/22

This is the final result for the 2018 Pb+Pb 5.02 TeV/n data. The uncertainty is **2.0%** if the data from runs 367318-367384, where the ATLAS toroid magnet is off, is included (i.e. using the HeavyIonP_ignore_TOROIDSTATUS.xml GRL) , or **1.9%** if this data is excluded (i.e. using the HeavyIonP_2018.xml GRL).

The following sentences can be written in public CONF notes/papers: "The uncertainty in the integrated luminosity of the data sample is 2.0\% (or 1.9\%). It is derived from the calibration of the luminosity scale using $\sqrt{x}\sqrt{y}$ beam-separation scans, following a methodology similar to that detailed in Ref.~\cite{DAPR-2013-01}, and using the LUCID-2 detector for the baseline luminosity measurements \cite{LUCID2}."

If this dataset is combined with the 2015 Pb+Pb dataset using the luminosity tag **OfiLumi-HI-010**, the uncertainty on the integrated luminosity of the combined dataset is 1.6% if the full dataset including the 2018 toroid-off data is used, and 1.5% if the toroid-off data is excluded. For the 2015 data, either tag OfiLumi-HI-007, OfiLumi-HI-009 or OfiLumi-HI-010 may be used (the tag contents for the 2015 data are identical). Details of the combination are shown [here](#), and the last slide of the talk summarises all the dataset combinations and possible uncertainties. The same sentence as above may be used in CONF notes/papers, quoting the 1.6% or 1.5% uncertainty in place of 2.0%.

See above/below for the references and BibTex entries.

2015 and 2017 5.02 TeV proton-proton (final)

Data Sample	data15/17_5TeV
Luminosity Tag	OfiLumi-5TeV-004
Luminosity Uncertainty	1.0% (combined), 1.0% (2017 only), 1.9% (2015 only, as previously)
Documentation	sigma_vis calibration , Calibration transfer, long-term stability and combination , Tag preparation
Last update for this entry	2 March 2023

The final calibration of 2017 5TeV pp reference data uses a calibration based on the vdM scans from 11-12 November 2017 (run 340634), a calibration transfer determined in the long reference run from 13-14 Nov 2017 (run 340718) which resulted in a scaling by 0.9983 applied to LUCID (no μ -dependence). The calibration transfer is based on track counting data which is anchored to LUCID in the head-on parts of vdM fill above. The default luminometer from the LUCID side is LUCID Bi EvtOR.

The updated tag is equivalent for 2017 to scaling the integrated luminosity by 0.9927. No change is done for 2015. Analyses can check the integrated luminosity of their GRL using the [LumiCalc tool](#). Using the standard GRL (see [GoodRunListsForAnalysisRun2#2017_low_mu_runs_summary](#)), you should get from the LumiCalc tool, the following numbers with the new luminosity tag:

For the application in the analysis of 2017 5TeV data, it is sufficient to rescale your MC samples with the factor 0.9927 to the new integrated luminosity, and rerun with the updated luminosity uncertainty. Additional ATLAS-internal information on the details of the calibration can also be found from the notes listed here: [LuminosityInternalNotes#2017_data](#)

The following sentence can be written in public CONF notes/papers for 2017 only, or for the combination of 2015 and 2017 data: "The uncertainty in the integrated luminosity is 1.0\%, following the methodology discussed in Ref.~\cite{DAPR-2021-01}, using the LUCID-2 detector~\cite{LUCID2} for the primary luminosity measurements, complemented by measurements using the inner detector and calorimeters."

Please quote as follows:

- In figure legends and in text if the uncertainty is not quoted: '255 pb⁻¹' (2017), '280 pb⁻¹' (2015+2017)
- If the absolute uncertainty is meant to be quoted: '254.9 $\pm$ 2.6 pb⁻¹' (2017), '280.1 $\pm$ 2.9 pb⁻¹' (2015+2017), following PDG rounding recommendations

The final 2015 calibration is discussed in a separate entry: [LuminosityForPhysics#2015_5_02_TeV_proton_proton_fina](#). For the references please see [LuminosityForPhysics#2015_2018_13_TeV_proton_proton_f](#).

2016 8.16 TeV proton-lead final January 2019

Data Sample	data16_hip8TeV
Luminosity Tag	OfiLumi-HI-006
Lumi Uncertainty	2.9/2.4/2.4% (p+Pb/Pb+p/combined)
Documentation	Slides Offline lumi 20/12/18

This is the final result for the 2016 p+Pb and Pb+p 8.16 TeV/n data. There are separate uncertainties for p+Pb (2.9%) and Pb+p (2.4%) data-taking periods, which have 1/3 and 2/3 of the integrated luminosity. The uncertainties are not fully correlated, as separate vdM scans were used to calibrate the two periods, and the combined uncertainty for the complete data sample (assuming all events are weighted equally) is **2.4%**.

The following sentences can be written in public CONF notes/papers: "The uncertainty in the integrated luminosity of the combined data sample is 2.4%. It is derived from the calibration of the luminosity scale using $\sqrt{s}$ - $\sqrt{s}$ beam-separation scans, following a methodology similar to that detailed in Ref.~\cite{DAPR-2013-01}, and using the LUCID-2 detector for the baseline luminosity measurements \cite{LUCID2}."

[DAPR-2013-01] ATLAS Collaboration, Luminosity Determination in pp Collisions at $\sqrt{s}$ = 8 TeV using the ATLAS Detector at the LHC, Eur. Phys. J. C 76 (2016) 653, arXiv:1608.03953 [hep-ex].

[LUCID2] G. Avoni et al., The new LUCID-2 detector for luminosity measurement and monitoring in ATLAS, JINST(2018) 13 P07017.

See above for the Bibtex entry for the LUCID paper.

[This entry last updated by RH on 23 Jan 2019]

2015 5.0 TeV lead-lead final

Data Sample	data15_hi
Luminosity Tag	OfiLumi-HI-009
Lumi Uncertainty	1.5%
Documentation	ATL-COM-DAPR-2019-022

This is the final result for the 2015 5.0 TeV/nucleon lead-lead data. The following sentence can be written in public CONF notes/papers: "The uncertainty in the integrated luminosity for the 2015 PbPb run at $\sqrt{s_{NN}}$ = 5.0 TeV is 1.5%. It is derived from the calibration of the luminosity scale using $\sqrt{s}$ - $\sqrt{s}$ beam-separation scans, following a methodology similar to that detailed in Ref.~\cite{DAPR-2013-01}, and using the LUCID-2 detector for the baseline luminosity measurements \cite{LUCID2}."

[DAPR-2013-011] ATLAS Collaboration, Luminosity Determination in pp Collisions at $\sqrt{s}$ = 8 TeV using the ATLAS Detector at the LHC, submitted to Eur. Phys. J. C, arXiv:1608.03953 [hep-ex].

2013 proton-lead final

Data Sample	data13_hip
Luminosity Tag	OfiLumi-HI-003
Lumi Uncertainty	2.7%
Documentation	ATL-COM-DAPR-2014-001 (v. 6.0)

This is the final result for the 2013 5.02 TeV proton-lead data. The total integrated luminosity went up by 1.03% from the preliminary result available at QM2014, and the uncertainty has been reduced from 3.4% to 2.7%. It is determined, following a methodology similar to that detailed in the 2011 luminosity paper (DAPR-2011-01, Eur. Phys. J. C (2013) 73:2518), from a calibration of the luminosity scale derived from beam-separation scans performed during the pPb operation of the LHC in 2013. The dominant uncertainty is associated with non-factorization effects.

2013 2.76 TeV proton-proton final

Data Sample	data13_2p76TeV
Luminosity Tag	OfiLumi-2p76TeV-002
Lumi Uncertainty	3.1%
Documentation	Internal Note

This is the final result for the 2013 2.76 TeV proton-proton data. The central value is unchanged from the preliminary result available at QM2014, but the uncertainty has been reduced from 3.4% to 3.1%. It is determined, following the same methodology as that detailed in the 2011 luminosity paper (DAPR-2011-01, Eur. Phys. J. C (2013) 73:2518), from a calibration of the luminosity scale derived from beam-separation scans performed during the 2.76 TeV operation of the LHC in 2013.

2011 2.76 TeV proton-proton final

Data Sample	data11_2p76TeV
Luminosity Tag	OfiLumi-2p76TeV-002
Lumi Uncertainty	2.7%
Documentation	Internal Note

The best reference for this is probably the [inclusive jet cross-section paper](#) which was the first publication to use this result.

2011 lead-lead (2.76 TeV /nucleon)

No approved vdM-based luminosity calibration exists for this running period.

2010 lead-lead (2.76 TeV /nucleon)

No vdM-based luminosity calibration exists for this running period.

ALFA data

2018 13 TeV proton-proton beta*=90m (preliminary)

Data sample	data18_13TeV (ALFA runs)
Luminosity Tag	Individually calibrated runs
Luminosity Uncertainty	1.0%
Documentation	vdM calibration , calibration transfer and long term stability (part 1) , calibration transfer and long term stability (part 2)
Last update for this entry	9 March 2023

The preliminary calibration of 2018 ALFA pp data (beta*=90m) uses the final calibration based on the vdM scans recorded 3-4 July 2018 (run 354494), and a dedicated calibration transfer and long term stability estimate, including the TightLumi and TightModLumi working points for track counting. The default luminometer from the LUCID side is LUCID EventORA Bi2. The uncertainty is 0.93% from the absolute calibration, 0.39% from the stability and consistency of luminosity algorithms, 0.041% for the background subtraction systematics, and 0.074% for beam gas, so together and rounded 1.0%.

The values for the integrated luminosities for the 4 runs of the ALFA 2018 13TeV period can be found:

- Per LB: [LumiPerLB_ALFA_90m.txt](#)
- Per Run: Slide 14 in [this presentation](#)
- Total: Slide 15 in the same presentation above

Please quote as follows:

- In figure legends and in text if the uncertainty is not quoted: '664 nb⁻¹'
- If the absolute uncertainty is meant to be quoted: '664 ± 7 nb⁻¹', following PDG rounding recommendations

Please contact the luminosity conveners before any publication including these values is released, to confirm that the central value and uncertainty are not in the process of being updated with improved calibration.

2016 13 TeV proton-proton beta*=2.5km (preliminary)

Data sample	data16_13TeV (ALFA runs)
Luminosity Tag	Individually calibrated runs
Luminosity Uncertainty	2.15%
Documentation	vdM calibration , calibration transfer and long term stability
Last update for this entry	6 February 2023

The preliminary calibration of 2016 ALFA pp data (beta*=2.5km) uses the preliminary calibration based on the vdM scans recorded 18-19 May and 27 May 2016 (runs 299390 and 300287), discussed in [ATLAS-CONF-2019-021](#), and a dedicated calibration transfer and long term stability estimate, including BCM and the TightModLumi working point from track counting, discussed in [STDM-2018-08](#). The default

luminometer from the LUCID side is LUCID EventORC Bi. The uncertainty is 1.1% from the absolute calibration and 1.85% from the stability and consistency of luminosity algorithms, so together 2.15%.

The final value of the integrated luminosity is: $L_{\text{int}} = 339.9 \pm 0.1 \text{ (stat.)} \pm 7.3 \text{ (syst.) } \mu\text{b}^{-1}$

Major updates:

-- [GiuliaRipellino](#) - 2026-06-24

Responsible: Antonio Sbrizzi, Giulia Ripellino

Last reviewed by: **Never reviewed**

Topic revision: r130 - 2026-07-06 - [KlausMoenig](#)

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